

```

1 .SUBCKT RUyyO_400k_120k_300MVA 1U 1V 1W 1N 2U 2V 2W 2N
2 + PARAMS:
3 + SN=300e6
4 + U1Nenn=400e3
5 + U20=120e3
6 + fn={fn1}
7 + N1=1000
8 + uR=0.002
9 + uX=0.2
10 + i0=0.0006
11 + p0=0.0004
12 + a=0.75 ; FS
13 + m=9 ; FS
14 + psi_0=0 ; Remanence-parameter 0,1,-1
15 + psir=1
16 + psis=-0.5
17 + psit=-0.5
18 + tpsi0=1000s ; Remanence-timepoint
19 + tpsid=0.1ms ; On-time of remanent flux
20 + ron=0.00001 ; On - Parameter of switcher for remanent flux
21 + bf=0 ; bf=1 -> middle leg with shorter magn. path correction factor
22 + rkessel=6e5 ; Tank resistance
23 + l_=0
24 + middle=0
25 + stproz=1.0
26 + ctrl={1+(l_-middle)*stproz/100} ; simple Tap changer
27 Xu 1U 1N 2U 2N magn1 magn2 Trafoschenkel2_Yy
28 + PARAMS:
29 + SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
30 + uX={uX} i0={i0*IF(bf==0,1,1.2)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
31 + psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
32 Xv 1V1N 2V 2N magn1 magn2 Trafoschenkel2_Yy
33 + PARAMS:
34 + SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
35 + uX={uX} i0={i0*IF(bf==0,1,0.6)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
36 + psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
37 Xw 1W 1N 2W 2N magn1 magn2 Trafoschenkel2_Yy
38 + PARAMS:
39 + SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
40 + uX={uX} i0={i0*IF(bf==0,1,1.2)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
41 + psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
42 Rm magn1 0 1MEG
43 Rkessel magn1 magn2 {rkessel}
44 RGND1U 1U 0 1e12
45 RGND1V 1V 0 1e12
46 RGND1W 1W 0 1e12
47 RGND2U 2U 0 1e12
48 RGND2V 2V 0 1e12
49 RGND2W 2W 0 1e12
50 RGND1N 1N 0 1e12
51 RGND2N 2N 0 1e12
52 .ENDS
53
54 .SUBCKT Trafoschenkel2_Yy 1_1 1_2 2_1 2_2 magnP1 magnP2
55 + PARAMS:
56 + SN=10kVA
57 + U1N=400V

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